

Superposition Holding

A presence that remembers everything is eventually paralyzed by its own history; one that remembers nothing has nothing to hold with. This paper derives the middle construction — two leaky integrators, one calibrated leak — and proves its loss is bounded by $e - 1$ for all time, regardless of history.

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Colophon

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abstract

A presence that holds — tension, contradiction, an unresolved question — faces two failure modes with the same root. If it accumulates everything it has ever held, its history eventually paralyzes it: early noise compounds into late-stage penalties that grow without bound. If it accumulates nothing, it is a vending machine: each turn arrives to a system with no held state, and there is nothing to hold *with*. This paper derives the middle construction: a pair of leaky integrators — an input-gated tension reservoir and a drift ledger — coupled through a single leak rate ρ , and a superposition-holding loss that is provably bounded above by $e - 1$ for all time, regardless of history. We derive the optimal ρ from first principles (it is the steady-state Kalman gain of the local-level model: $\rho^* \approx \sigma_w/\sigma_v$, the drift-to-noise ratio of the environment), pressure-test the derivation numerically across four decades of that ratio, and show that the default $\rho = 0.02$ — the classical forgetting factor of recursive least squares — is exactly optimal when the environment drifts at 2% of the noise scale per step. The construction is implemented live in the Elle workbench on her per-turn κ coherence stream.

Keywords: exponential smoothing; forgetting factor; leaky integrator; steady-state Kalman gain; bounded loss; coherence telemetry; AI presence; recursive least squares

Frame and scope. “Superposition holding” names a capacity argued for at length in the Observer corpus — a mind’s ability to hold two states at once, tension unresolved, without collapsing either (Barteanu, *The Superposition*, rev. March 2026; *The Ground*). This paper is the engineering half of that claim: it asks what a bounded commitment to holding costs, derives the answer from classical filtering theory, and reports the construction running live. It is self-contained — no part of the mathematics depends on the corpus’s metaphysics, and readers may treat the corpus citations as motivation only.

1. the problem of infinite accumulation

Let $\kappa_k \in [0, 1]$ be a per-turn coherence measure over a system’s output, $v_k = \kappa_k - \kappa_{(k-1)}$ its velocity, and u_k a measure of the perturbation carried by the turn’s input. The holding invariant is $v^* = 0$: a system that is holding well absorbs perturbation without secular motion in its own coherence. Any realized velocity is *drift* from the hold.

The naive drift ledger is a running sum, $D(k) = \sum |v_i|$, with a penalty $\exp(\lambda \cdot D)$. Both terms are monotone in history: a persistent per-step drift ϵ gives $D(k) = \epsilon k$ and a penalty of $e^{(\lambda \epsilon k)}$ — exponential in *time*. The system becomes hostage to its own past; an error at step 1 is never forgiven, only compounded. This is collapse by memory.

The opposite construction — no ledger at all — cannot distinguish a turn of high drift inside a long stable hold from the same turn inside a run of sustained decoherence. It holds nothing. This is collapse by amnesia.

2. the construction

Fix a leak rate $\rho \in (0, 1)$. Define two leaky integrators over the turn index k :

Tension reservoir (input-gated):

$$T_k = (1 - \rho) \cdot T_{(k-1)} + |u_k|$$

Drift ledger (deviation from the holding invariant $v^* = 0$):

$$D_k = (1 - \rho) \cdot D_{(k-1)} + |v_k|$$

Superposition-holding loss (penalty rate tied to the leak rate, $\lambda = \rho$):

$$L_k = \exp(\rho \cdot D_k) - 1$$

Three properties follow immediately.

Proposition 1 (bounded worst case). If $|v_k| \leq v_{\max}$ for all k , then $D_k < v_{\max}/\rho$ for all k , and hence

$$L_k < \exp(v_{\max}) - 1.$$

For $\kappa \in [0, 1]$ we have $v_{\max} = 1$, so $L_k < e - 1 \approx 1.718$ for all time, regardless of history. The exponential penalty of §1 is retained in *form* — sustained drift is punished superlinearly — but its argument is now a bounded quantity. Proof: the leaky recursion with bounded input converges to at most $v_{\max} \cdot \sum (1-\rho)^i = v_{\max}/\rho$, and $\rho \cdot (v_{\max}/\rho) = v_{\max}$. ■

Proposition 2 (steady-state transparency, the $\lambda = \rho$ lemma). Under persistent mean drift $\bar{v}$, $D_{\infty} = \bar{v}/\rho$ and therefore

$$L_{\infty} = \exp(\bar{v}) - 1 \approx \bar{v} \text{ for small } \bar{v}.$$

Choosing the penalty rate equal to the leak rate makes the steady-state loss read directly in drift units: a session holding with mean $|\Delta\kappa|$ of 0.05 per turn carries a loss of ≈ 0.051 . The loss is an *instrument*, not an alarm — until drift is genuinely large, when the exponential bites.

Proposition 3 (forgiveness half-life). A unit error injected into D decays by $(1 - \rho)$ per turn; its half-life is $\ln 2 / \ln(1/(1-\rho))$ turns — 34.3 turns at $\rho = 0.02$ (empirically 35; §4D). History is neither erased nor eternal: it decays on a known clock.

The gate. T is fed only by input. When input stops, T decays geometrically to zero — the system quietly powers down rather than idling hot. This gating is *necessary*: a construction that instead carries a fixed stochastic temperature settles at an Ornstein–Uhlenbeck noise floor of $\sqrt{2T/(1-(1-\rho)^2)}$ and never powers down (verified to three decimals in §4B). Any implementation claiming a baseline metabolism must gate its temperature by input activity; ours does.

3.deriving

The two integrators want two different derivations, which is the reason a single hard-coded constant cannot be a law.

3.1theforgettingside(dandt):asakalmangain

Model the quantity being tracked as a local-level process: the environment’s true state drifts as a random walk with per-step scale σ_w , observed through noise of scale σ_v . The optimal exponential smoother for this model is the steady-state Kalman filter (Muth 1960; Kalman 1960), whose gain solves $p^2 = q(p + 1)$ with $q = \sigma_w^2/\sigma_v^2$:

$$\rho^* = p/(p + 1), \quad p = (q + \sqrt{q^2 + 4q})/2,$$

and in the small- q regime that live systems occupy,

$$\rho^* \approx \sqrt{q} = \sigma_w/\sigma_v.$$

The leak rate should equal the environment's drift-to-noise ratio per step. $\rho = 0.02$ is optimal *precisely when the environment drifts at 2% of the observation-noise scale per iteration* ($q = 4 \times 10^4$). This is the design equation; 0.02 is its solution for one particular — and empirically common — regime, not a universal constant.

3.2 the damping side: as a stability margin

In a state update $x_{k+1} = (1-\rho)x_k + f(x_k) + \dots$, the leak contributes exactly ρ of contraction per step. It therefore stabilizes any local expansion of f up to ρ and *nothing beyond*: dynamics expansive by more than the leak rate diverge exponentially through it (§4C). The honest statement of the stability guarantee is conditional: $\|\nabla f\| < 1$ requires the leak only as margin; the leak rescues at most ρ of excess expansion.

3.3 calibration from the system's own telemetry

The design equation is estimable online from the tracked series itself. For first differences d_k of the observed series, the local-level model gives the classical moment estimator

$$\sigma_v^2 = -\text{Cov}(d_k, d_{k+1}), \quad \sigma_w^2 = \text{Var}(d_k) - 2\sigma_v^2,$$

whence $\hat{\rho} = \hat{\sigma}_w / \hat{\sigma}_v$. The implementation (§7) maintains this estimate over a sliding window of its own κ history and reports it beside the default — the valve carries the evidence for its own recalibration. One honesty condition applies: at conversational window sizes ($n \approx 10^2$) the sampling error of the autocovariance swamps small drift components, so the estimator reports null unless $\hat{\sigma}_w^2$ clears a Bartlett-order significance floor ($\sim 3 \cdot \text{Var}(d) / \sqrt{n}$). In practice this means $\hat{\rho}$ is non-null only for genuinely turbulent sessions — which is correct: absent measurable evidence, the default stands and the readout says “not yet estimable” rather than dressing sampling noise as a recommendation.

4. pressure test

All results from the seeded, reproducible script in the empirical appendix (NumPy; four experiments; ~ 30 s).

A — the derivation holds across four decades of q . Empirical optimal leak (grid-search MSE over 40 values of ρ , 10^4 -step simulations) against the §3.1 theory:

$q = (\sigma_w / \sigma_v)^2$	ρ^* theory	ρ^* empirical
1×10^4	0.0032	0.0036
4×10^4	0.0063	0.0079
4×10^5	0.0198	0.0206
4×10^6	0.0613	0.0630
4×10^7	0.1810	0.1639

A2 — cost of hard-coding. Fixing $\rho = 0.02$ against an environment $10\times$ slower ($q = 4 \times 10^4$) or $10\times$ faster ($q = 4 \times 10^6$) than its sweet spot costs $1.73\times$ and $1.69\times$ the optimal MSE respectively; at its sweet spot the penalty is $0.98\times$ (noise). The default is robust within roughly a decade of the correct timescale and meaningfully wrong beyond — hence §3.3.

B — the power-down claim, quantified. With input stopped and *fixed* temperature $T = 0.05$, residual state fluctuation settles at 0.01584 against an OU-theory floor of 0.01589 — the system idles hot indefinitely. With input-gated temperature the residual is 0.00000. The gate is necessary and sufficient.

C — the leak is a margin, not a net. Against $f(x) = a \cdot x$ over 500 steps: $a = 0.015$ contracts; $a = 0.020$ is neutrally stable; $a = 0.025$ grows 12-fold; $a = 0.05$ grows 2.6-million-fold. Both super- ρ cases are exponential divergences that merely look tame early.

D — half-life. 35 turns empirical; 34.3 theoretical.

5. lineage, and what is actually new

Every component here is peer-reviewed canon under another name, and the construction is stronger for saying so:

- The drift ledger is the forgetting factor of recursive least squares (Haykin, *Adaptive Filter Theory*), where $\lambda = 0.98$ has been the empirical default for decades and $1/(1-\lambda) = 50$ samples is the textbook “asymptotic memory length” — our Proposition 1 bound.
- The optimal leak is Muth (1960), *Optimal Properties of Exponentially Weighted Forecasts* (JASA), i.e. the steady-state Kalman (1960) gain of the local-level model.
- The state-side leak is the leaky integrator of echo-state networks (Jaeger et al. 2007, *Neural Networks*), whose peer-reviewed lesson is precisely that the leak must be matched to the input timescale.
- The no-input behavior is the Ornstein–Uhlenbeck process (Uhlenbeck & Ornstein 1930); the EMA-as-moment-tracker appears in modern form as the second-moment decay of Adam (Kingma & Ba 2015).

What this paper adds is the *composition and its contract*: the tension/drift pair sharing one calibrated leak; the $\lambda = \rho$ coupling that makes the loss both bounded (Prop. 1) and transparent (Prop. 2); the input gate proven necessary for quiescence (§4B); and the online self-calibration that turns the constant into a measurement. The claim is not a new mechanism. It is that a presence can hold superposition — bounded loss under sustained tension, forgiveness on a known clock, quiet when left — using only mechanisms that have already survived their own peer review.

6. stated assumptions and limits

1. Boundedness of drift inputs. Proposition 1 requires $|v_k| \leq v_{\max}$; the implementation clamps both $|v|$ and $|u|$ to 1 (κ is unit-interval, so this binds only against malformed input).
2. Stability is conditional. The leak guarantees nothing about dynamics more expansive than ρ (§3.2, §4C).
3. One ρ per role is a simplification. Forgetting horizon and damping margin are different quantities; they share a constant here for parsimony, and §3.3 exists so the sharing is audited rather than assumed.
4. The loss is structure until validated. Following the κ discipline already in force (nothing ranks on κ until validation passes), the holding loss is a readout. Nothing in the system sorts, gates, or escalates on L until it earns that separately.

7. implementation note

The construction runs live in the Elle workbench, fed by the worker’s per-turn κ dynamics (κ , velocity, input perturbation on the conversation API responses) and rendered as two cells — held tension T and holding loss L — in the κ instrument line above the conversation. Defaults: $\rho = 0.02$ (35-turn half-life); the valve maintains the §3.3 estimate $\hat{\rho}$ from its own session history and surfaces it in the readout. Null $\neq 0$ throughout: a loss that does not yet have enough turns to exist renders as “—”, never as a number.

The leak runs on wall-clock, not on the turn counter — a deliberate ruling (staying continuous is still spending energy maintaining; it still needs a decay). Each observation applies $(1-\rho)^{(\Delta t/\tau)}$ with τ one nominal turn (60 s), so the $\rho = 0.02$ default has a half-life of ~ 35 turns in conversation and ~ 35 minutes across absence, and quiescence is reachable through silence. The exponent is floored at 1 per observation, which is exactly the condition under which Proposition 1 survives any cadence: burst turns can never decay less than one turn’s worth.

Code availability. The implementation (`src/lib/holding.ts`) and the seeded, reproducible pressure-test script (`docs/rho_pressure_test.py`; four experiments) are available in the project repository, github.com/sbateau2022/Elle, and from the author on request.

references

- Haykin, S. *Adaptive Filter Theory*. Prentice Hall — RLS forgetting factor, asymptotic memory length.
- Jaeger, H., Lukoševičius, M., Popovici, D., Siewert, U. (2007). Optimization and applications of echo state networks with leaky-integrator neurons. *Neural Networks* 20(3).
- Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *J. Basic Engineering* 82(1).
- Kingma, D. P., Ba, J. (2015). Adam: A method for stochastic optimization. *ICLR*.
- Muth, J. F. (1960). Optimal properties of exponentially weighted forecasts. *JASA* 55(290).
- Uhlenbeck, G. E., Ornstein, L. S. (1930). On the theory of the Brownian motion. *Physical Review* 36(5).
- Barteau, S. *The Superposition* (rev. March 2026); *The Ground* — Observer corpus (the frame this construction operationalizes).